

QVerse — Week 4: Relative Phase and Interference

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

2026

How this booklet fits the course

This theory booklet is designed to be read alongside `04_phase_and_interference.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 4; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Explain constructive and destructive interference.
- Show how relative phase becomes observable.
- Use H-Z-H and H-X-H identities.
- Explore S and T phase gates.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Why interference is central

Many quantum algorithms are not useful merely because they create superpositions. They are useful because they manipulate **relative phases** and then use **interference** to suppress unwanted outcomes and enhance useful ones.

The correct mental model is not “quantum computers evaluate all answers at once.” A better model is:

1. amplitudes are distributed over many basis states;
2. phases are modified according to the problem;
3. amplitudes are recombined;
4. interference changes the final measurement probabilities.

The measurement only reveals the result of this engineered interference pattern.

2. Amplitudes add before probabilities are computed

Suppose two computational paths contribute amplitudes a_1 and a_2 to the same final state. The total amplitude is

$$a_{\text{tot}} = a_1 + a_2.$$

The final probability is

$$P = |a_1 + a_2|^2.$$

It is generally **not**

$$|a_1|^2 + |a_2|^2.$$

The cross terms in $|a_1 + a_2|^2$ produce interference.

If $a_1 = a_2$, the amplitudes reinforce. If $a_1 = -a_2$, they cancel.

This is why phase matters even when phase does not directly appear in a computational-basis histogram.

3. Relative phase

For

$$|\psi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}},$$

the computational-basis probabilities are

$$P(0) = P(1) = \frac{1}{2}$$

for every ϕ .

So a direct Z-basis measurement cannot determine ϕ .

But after another basis-changing gate, the phase changes the amplitudes that interfere and therefore changes the probabilities. Relative phase is hidden information that can become visible.

4. H-Z-H as an interference experiment

Start from

$$|0\rangle.$$

After the first Hadamard,

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Apply Z:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{Z} \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

Then apply H:

$$|-\rangle \xrightarrow{H} |1\rangle.$$

Thus

$$HZH|0\rangle = |1\rangle.$$

The middle Z gate never changed the 50/50 Z-basis probabilities at the intermediate point. It changed a phase, and the final H converted that phase difference into a deterministic population difference.

This is the prototype of many larger quantum algorithms.

5. H-X-H

Similarly,

$$HXH = Z.$$

The Hadamard changes the basis in which the middle operation should be interpreted.

The two identities

$$HZH = X, \quad HXH = Z$$

show that H interchanges the roles of bit flips and phase flips.

This idea later appears in error correction, teleportation, and algorithm design.

6. S and T phase gates

The S gate is

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

It changes the relative phase of the $|1\rangle$ component by $\pi/2$.

The T gate is

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

It changes the relative phase by $\pi/4$.

Applied to $|+\rangle$,

$$S|+\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}},$$

and

$$T|+\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}.$$

The immediate Z-basis probabilities remain 50/50. The difference becomes visible only after an appropriate interference operation.

7. The continuous phase gate $R_z(\phi)$

A standard z -axis rotation is

$$R_z(\phi) = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}.$$

Up to global phase, it changes the relative phase between $|0\rangle$ and $|1\rangle$.

A very useful interference experiment is

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{R_z(\phi)} \text{phase-shifted state} \xrightarrow{H} \text{measurement}.$$

The final probability oscillates as the phase is varied. Depending on the exact convention,

$$P(1) = \sin^2\left(\frac{\phi}{2}\right)$$

for the standard $H - R_z(\phi) - H$ arrangement starting from $|0\rangle$.

This continuous oscillation is a direct way to see phase converted into measurable probability.

8. Global phase still does not matter

A gate sequence can differ from another by an overall multiplicative factor $e^{i\gamma}$. Such circuits are physically equivalent for ordinary measurement predictions.

When checking whether two unitaries are equivalent, distinguish:

- exact matrix equality;
- equality up to global phase.

Qiskit utilities such as `Operator.equiv` are designed to make this distinction.

9. Interference in algorithm design

A common quantum-algorithm architecture is:

prepare $\rightarrow$ phase encoding $\rightarrow$ interference $\rightarrow$ measure.

Examples later in the course:

- Deutsch-Jozsa uses the oracle to encode information into phase and Hadamards to interfere amplitudes.
- Grover uses a phase oracle and a diffuser to amplify marked-state amplitude.
- QFT reorganizes relative phases into a useful structured representation.
- variational circuits tune phases and rotations to minimize an expectation value.

Interference is therefore not an isolated one-qubit curiosity. It is one of the main resources used by quantum algorithms.

10. Constructive and destructive interference

If two amplitudes arrive with the same phase,

$$a + a = 2a,$$

so the probability scales as

$$|2a|^2 = 4|a|^2.$$

If they arrive with opposite phase,

$$a - a = 0,$$

the corresponding outcome disappears.

In a normalized quantum state, reinforcement in one part of the state is accompanied by reduced amplitude elsewhere. A unitary does not create total probability; it redistributes amplitudes while preserving normalization.

11. Why probabilities alone are insufficient

Suppose two states have identical computational-basis probabilities:

$$|\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Direct Z measurement sees no difference. But after applying H :

$$H|\psi_1\rangle = |0\rangle, \quad H|\psi_2\rangle = |1\rangle.$$

Therefore the full quantum state contains information that cannot be represented only by a list of basis-state probabilities.

12. Common misconceptions

“A phase gate does nothing because the histogram does not change”

A phase gate may leave one measurement basis unchanged while altering the outcome after later interference.

“Interference is just ordinary random cancellation”

Interference happens at the amplitude level, before probabilities are formed.

“All phases are observable”

Only relative phase matters. Overall global phase is not.

“Superposition alone gives computational advantage”

A useful algorithm must arrange interference so that the final measurement contains useful information.

13. What to know before the notebook

You should be able to explain:

- why amplitudes, not probabilities, interfere;
- the difference between global and relative phase;
- why $HZH = X$;
- what S , T , and R_z do conceptually;
- how a phase can be invisible before a basis change and visible afterward.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.